Polytrauma & Spinal Cord Injuries: : A Clinical Pharmacy Perspective

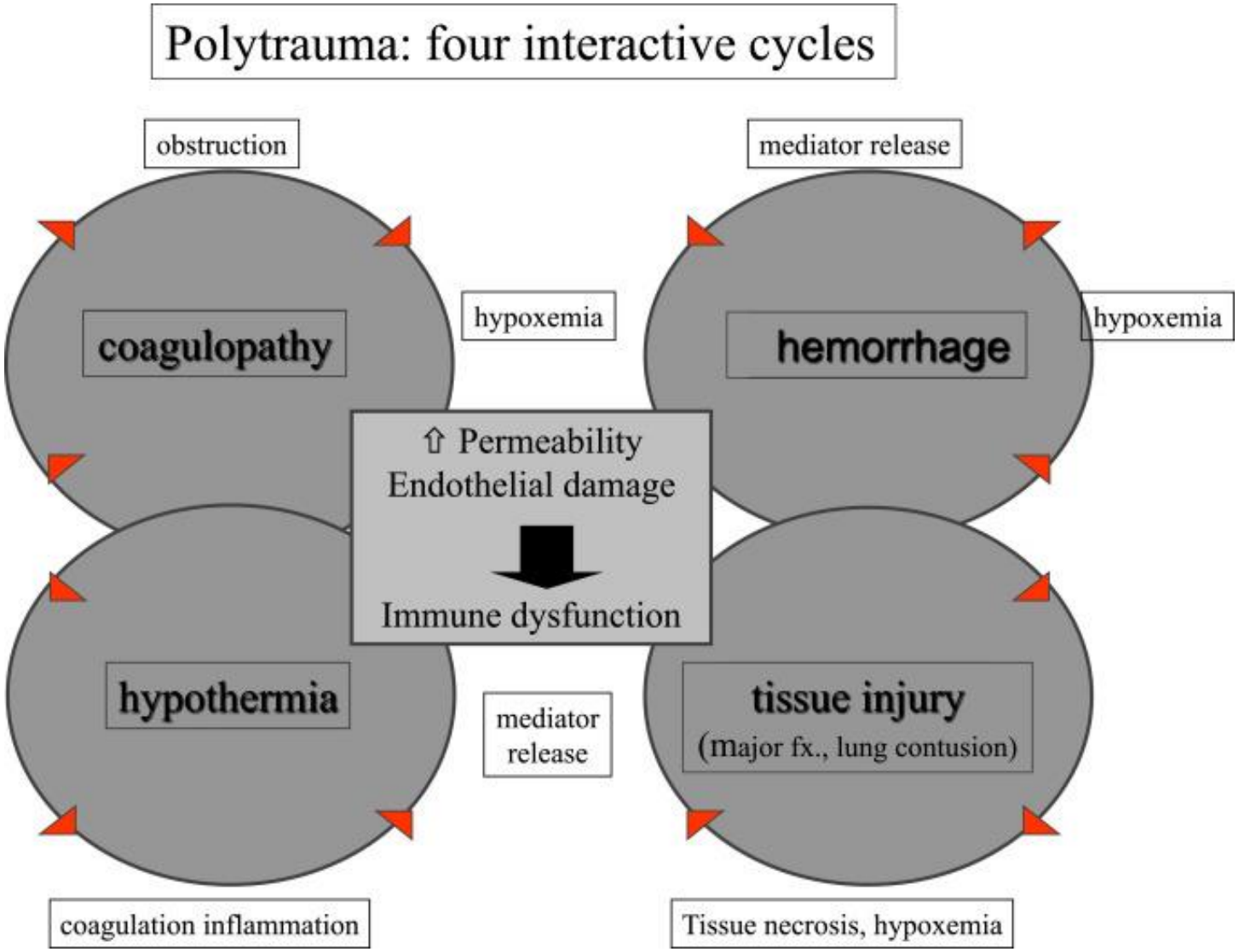
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Def. of Polytrauma:

- two or more injuries, one of which may be life threatening
- sustained in the same incident
- that affect multiple body parts or organ systems



Risks with polytrauma:

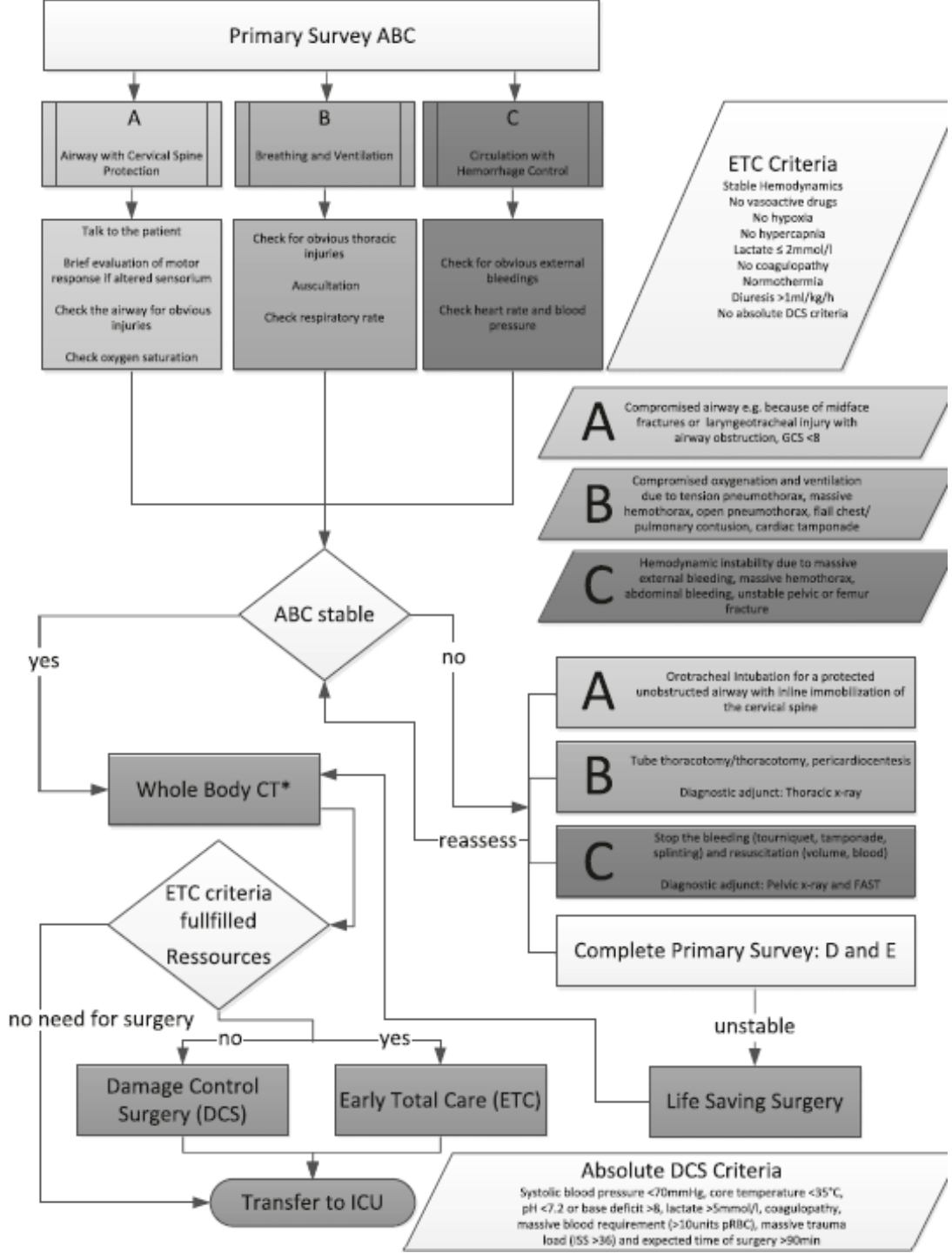
- Death:
  - Three peak times:
    - 50% within the first minutes of sustaining the injury
      - caused by massive blood loss or neurologic injury
    - 30% within hours of arrival to hospital
      - most commonly from shock, hypoxia, or neurologic injury
    - 20% within days to weeks following injury
      - multi system organ failure and infection are leading causes
- Physical, cognitive, psychological, or psychosocial impairments and functional disability.

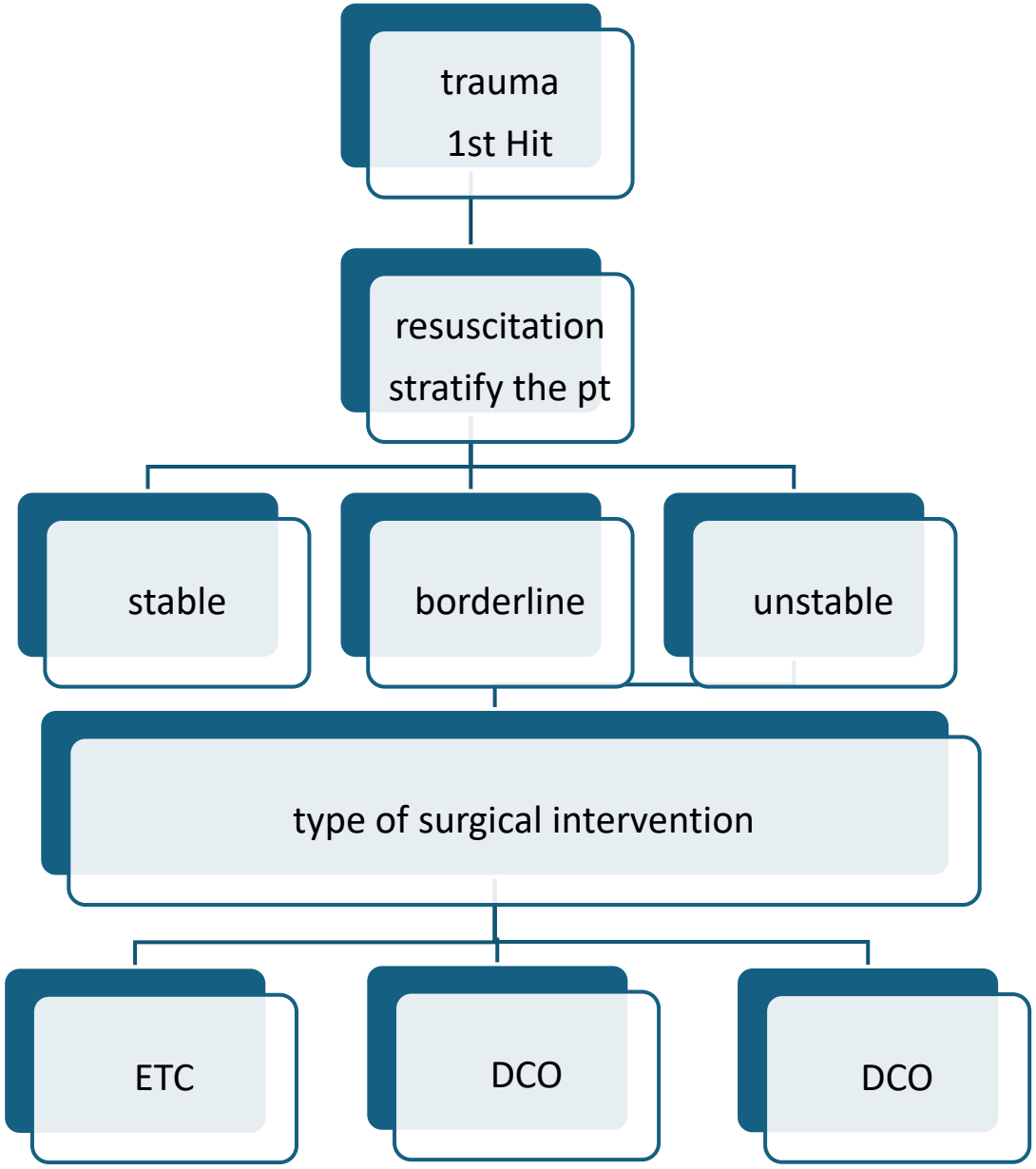
How to deal with a polytrauma patient ???

……Through **ATLS**

Advanced Trauma Life Support:

- A protocol for managing the polytrauma patient
- Incudes:
  - Primary survey (ABCDE)
    - treat greatest threats to life first = Resuscitation
    - includes cervical spine control
  - Secondary survey
    - AFTER life-threatening injuries have been addressed
    - Head to toe examination and updated history
    - obtain indicated imaging studies
  - Tertiary survey
    - repeat physical examination and additional imaging as indicated
  - Spine injuries are of paramount importance in polytrauma patients
    - Can cause spinal cord injuries
    - Disability
    - Death in high level


Primary Survey ABC
A
Airway with Cervical Spine Protection
B
Breathing and Ventilation
C
Circulation with Hemorrhage Control
Talk to the patient
Brief evaluation of motor response if altered sensorium
Check the airway for obvious injuries
Check oxygen saturation
Check for obvious thoracic injuries
Auscultation
Check respiratory rate
Check for obvious external bleedings
Check heart rate and blood pressure
ETC Criteria
Stable Hemodynamics
No vasoactive drugs
No hypoxia
No hypercapnia
Lactate ≤ 2mmol/l
No coagulopathy
Normothermia
Diuresis >1ml/kg/h
No absolute DCS criteria
A
Compromised airway e.g. because of midface fractures or laryngeotracheal injury with airway obstruction, GCS <8
B
Compromised oxygenation and ventilation due to tension pneumothorax, massive hemothorax, open pneumothorax, flail chest/ pulmonary contusion, cardiac tamponade
C
Hemodynamic instability due to massive external bleeding, massive hemothorax, abdominal bleeding, unstable pelvic or femur fracture
ABC stable
yes
no
A
Orotracheal intubation for a protected unobstructed airway with inline immobilization of the cervical spine
B
Tube thoracotomy/thoracotomy, pericardiocentesis
Diagnostic adjunct: Thoracic x-ray
C
Stop the bleeding (tourniquet, tamponade, splinting) and resuscitation (volume, blood)
Diagnostic adjunct: Pelvic x-ray and FAST
Complete Primary Survey: D and E
reassess
Whole Body CT*
ETC criteria fullfilled
Ressources
no need for surgery
no
yes
unstable
Damage Control Surgery (DCS)
Early Total Care (ETC)
Life Saving Surgery
Transfer to ICU
Absolute DCS Criteria
Systolic blood pressure <70mmHg, core temperature <35°C, pH <7.2 or base deficit >8, lactate >5mmol/l, coagulopathy, massive blood requirement (>10units pRBC), massive trauma load (ISS >36) and expected time of surgery >90min


trauma
1st Hit
resuscitation
stratify the pt
stable
borderline
unstable
type of surgical intervention
ETC
DCO
DCO

	Parameter	Stable (Grade I)	Borderline (Grade II)	Unstable (Grade III)	In Extremis (Grade IV)
Shock	Blood pressure (mm Hg)	100 or more	80–100	60–90	<50–60
	Blood units (2 h)	0–2	2–8	5–15	>15
	Lactate levels	Normal range	Around 2.5	>2.5	Severe acidosis
	Base deficit (mmol/L)	Normal range	No data	No data	>6–8
	ATLS classification	I	II–III	III–IV	IV
Coagulation	Platelet count (µg/mL)	>110	90–110	<70–90	<70
	Factor II and V (%)	90–100	70–80	50–70	<50
	Fibrinogen (g/dL)	1	Around 1	<1	DIC
	D-dimer	Normal range	Abnormal	Abnormal	DIC
Temperature		<33°C	33–35°C	30–32°C	30°C or less
Soft Tissue Injuries	Lung function; PaO_2/FiO_2	350–400	300–350	200–300	<200
	Chest trauma scores; AIS	AIS 1 or 2	AIS 2 or more	AIS 2 or more	AIS 3 or more
	Chest trauma score; TTS	0	I–II	II–III	IV
	Abdominal trauma (Moore)	< or = II	< or = III	III	III or > III
	Pelvic trauma (AO class.)	A type (AO)	B or C	C	C (crush, rollover abd.)
	Extremities	AIS I–II	AIS II–III	AIS III–IV	Crush, rollover extrem.

Table 2. Classification of Hemorrhagic Shock.*

Shock Class	Blood Loss†	Heart Rate	Blood Pressure	Pulse Pressure	Respiratory Rate	Mental Status
	ml (%)	*beats/min*			*breaths/min*	
I	<750 (15)	<100	Normal	Normal	14–20	Slightly anxious
II	750–1500 (15–30)	100–120	Normal	Narrowed	20–30	Mildly anxious
III	1500–2000 (30–40)	120–140	Decreased	Narrowed	30–40	Anxious, confused
IV	>2000 (>40)	>140	Decreased	Narrowed	>35	Confused, lethargic

* Data are from the American College of Surgeons Committee on Trauma.[42]
† Blood-loss volume and percentage of total blood volume are for a male patient with a body weight of 70 kg.

Definitions of ETC and DCO:

- Early Total Care:
  - the concept of fixing long bone fractures as soon as possible because patients were "too sick not to operate"
  - ETC led to exacerbation of the "second-hit" in a subset of patients with hemodynamic instability, head, and/or chest injuries
- Damage Control Orthopaedics (DCO)
  - definitive treatment delayed until physiology has improved
  - replaced the philosophy of Early Total Care (ETC)

Methods of resuscitation:

- **Crystalloids**
  - **First-line fluid for resuscitation.**
  - 2 L isotonic NaCl or lactated Ringer's solution as bolus initially.
  - Continue until hemodynamic stability is achieved.
  - 3:1 ratio, ie: 3 L of crystalloid required to raise intravascular volume by 1 L (due to vascular leak).
- **Colloids**
  - 1:1 ratio.
  - Options: Human albumin, hydroxy-ethyl starch, hypertonic saline-dextran.
  - Hypertonic saline-dextran: Useful in limited fluid scenarios (e.g., elderly with cardiac impairment).
- **Blood Transfusion**
  - **Indication:** Persistent instability after 2000 mL crystalloid.
  - Use O-negative noncrossmatched blood in acute cases (2 U rapidly).
  - Type and Cross: Draw sample before transfusion; switch to type-specific blood when available.
  - Risks: Coagulopathy with large transfusions.
- **Coagulopathy Management**
  - **1:1:1 ratio** (RBCs : PLTs : Plasma)
  - **FFP:** Infuse after 6-8 U PRBCs or signs of coagulopathy.
  - **Platelets:** Transfuse when depleted or coagulopathy develops.

Indicators of adequate resuscitation

- urine output 0.5-1.0 ml/kg/hr (30 cc/hr)
- serum lactate levels
  - normal < 2.5 mmol/L, < 45 mg/dL
  - most sensitive indicator as to whether some circulatory beds remain inadequately perfused
- base deficit
  - normal -2 to +2

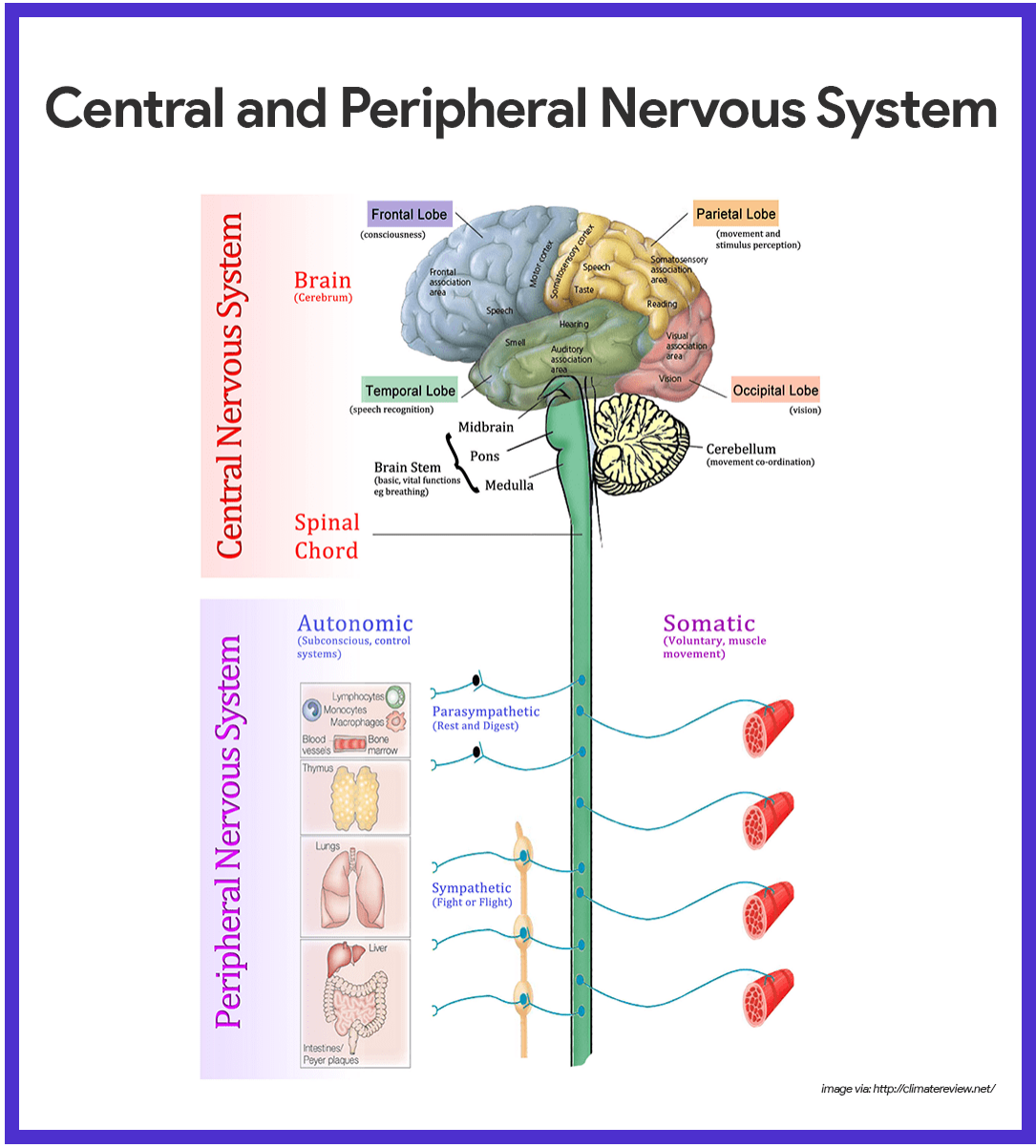
Spinal cord injuries

- Demographics
  - bimodal distribution
    - *young individuals* with significant trauma
    - *older individuals* that have minor trauma compounded by degenerative spinal canal narrowing
  - Anatomic location
    - 50% in cervical spine



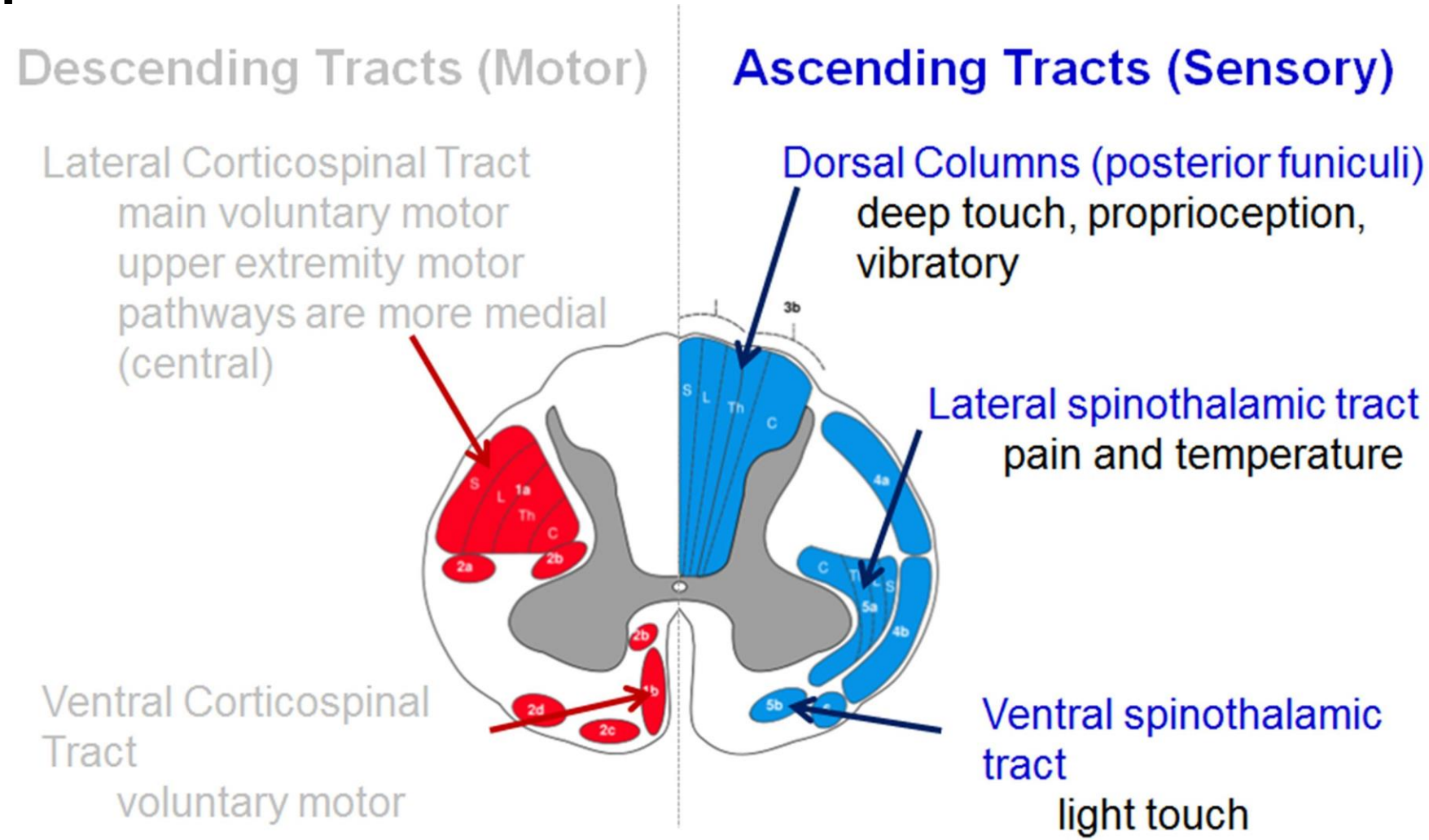
Anatomy of the nervous system:

- central nervous system
  - brain
  - spinal cord
- peripheral nervous system
  - cranial nerves
  - peripheral nerves
- autonomic nervous system
  - sympathetic system
    - a total of 22 ganglia (3 cervical, 11 thoracic, 4 lumbar, and 4 sacral)
  - parasympathetic nervous system
    - hypogastric plexus
    - formed by S2, S3, and S4 parasympathetic fibers and lumbar sympathetic fibers (splanchnic nerves)



Anatomy of spinal cord:

- It consists of tracts; collection of nerve fibers which transmit signals either upwards or downwards.

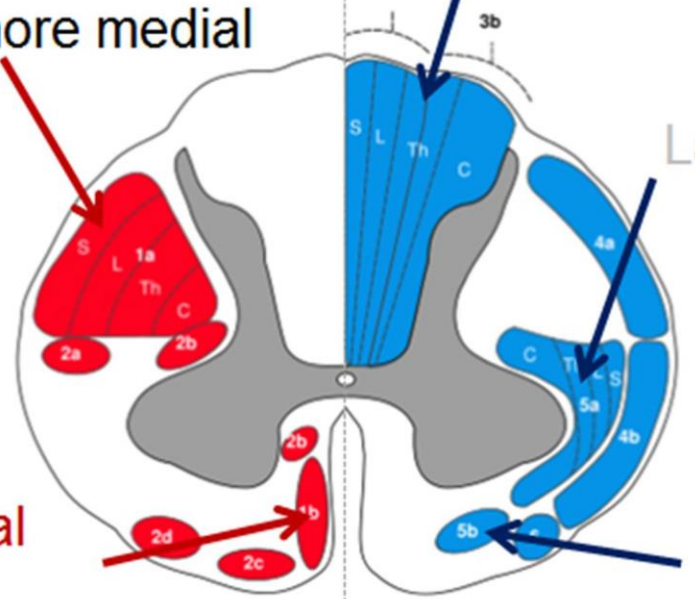


Descending Tracts (Motor)

Lateral Corticospinal Tract

main voluntary motor

upper extremity motor pathways are more medial (central)



Ventral Corticospinal Tract

voluntary motor

Ascending Tracts (Sensory)

Dorsal Columns (posterior funiculi)

deep touch, proprioception, vibratory

Lateral spinothalamic tract

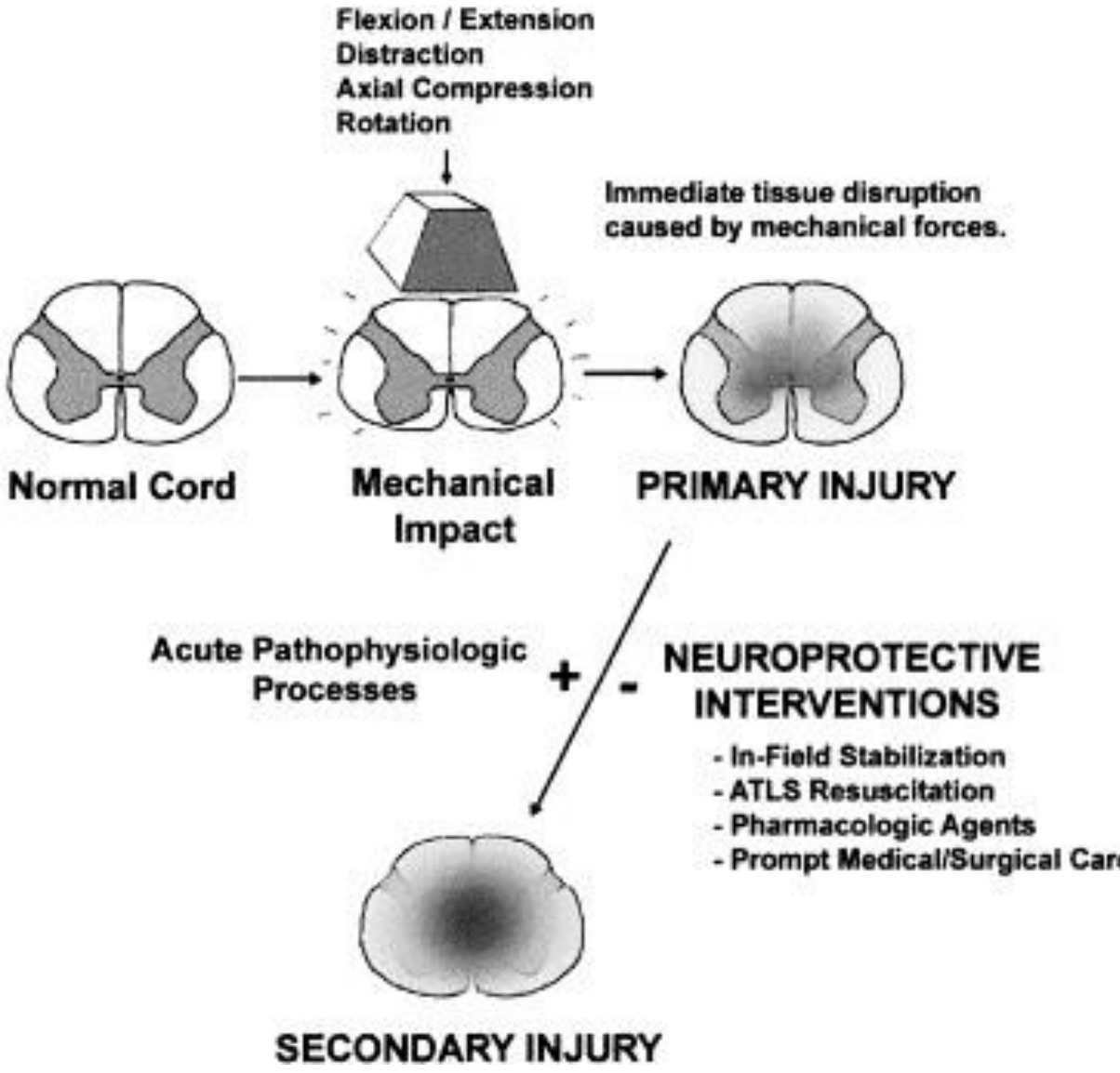
pain and temperature

Ventral spinothalamic tract

light touch

- Mechanism of spinal cord injuries:
  - MVA causes 50%
  - Falls, gunshot, …
- Pathophysiology
  - primary injury
    - damage to neural tissue due to direct trauma
    - irreversible
  - secondary injury
    - injury to adjacent tissue due to
      - decreased perfusion
      - lipid peroxidation
      - free radical / cytokines release
      - cell apoptosis
    - methylprednisolone used to prevent secondary injury ??

      → by improving perfusion, inhibiting lipid peroxidation, and decreasing the release of free radicals



classification

Quadriplegia	paraplegia	complete injury	incomplete injury
• injury to the cervical spinal cord • leading to impairment of function in the arms, trunk, legs, and pelvic organs	• injury to the thoracic, lumbar, or sacral segments • leading to impairment of function in the trunk, legs, and pelvic organs depending on the level of injury • arm function is preserved	• an injury with no spared motor or sensory function below the affected level	• an injury with some preserved motor or sensory function below the injury level

Medical ttt for patients with spinal cord injuries:

DVT prophylaxis

- modalities
  - low-molecular weight heparin
  - rotating bed
  - pneumatic compression stocking

Cerebrolysin (CBL)

- Peptide with neurotrophic and neuroprotective activity

Steroids

- high dose methylprednisolone
- Controversial now
- Load 30mg/kg in 1 hr
- Maintain 5.4mg/kg/hr
- Max 48 hrs

Urinary tract infections:

- Proper antibiotics + sterile catheterization techniques

Bowel management

- High Fibre Diet
- Adequate Fluid Intake
- Medication
  - Oral e.g. Bowel Softeners
  - Rectal e.g. Enemas

Sinus bradycardia

- most common cardiac arrhythmia in acute stage following SCI

Orthostatic hypotension

- occurs because of lack of sympathetic tone